cases is that a relatively small amount of activity may be isolated from a large mass of isotopic inactive target. Because of retention there is also some labeling of the target molecules.

## 17.7. FAST RADIOCHEMICAL SEPARATIONS

Short-lived radioactive nuclides are of interest (i) to nuclear science to establish the outer boundary for nuclide formation, which yields information on nuclear structure , (ii) to chemistry to extend the limits of the periodic system, i.e. the actinides and transactinides (or *superheavy elements*); these studies are also of interest to cosmology since these heaviest elements are believed to be formed in supernovae, and (iii) to medicine where they offer a unique tool to identify and treat some diseases (§18.5). Short-lived radionuclides are also used in chemical and industrial research, one of their advantages being that they rapidly disappear, eliminating radioactive waste problems. We arbitrarily define short-lived as meaning half-lives $<10$ minutes. Because many of the lightest elements of greatest interest in organic chemistry and biology lack long-lived radionuclides, the production of their short-lived isotopes are of particular interest; see Figure 5.7. At the other end of the periodic system there are no stable isotopes at all, and for the trans-uranium elements the half-lives of the most stable isotopes become shorter and shorter for each higher element, e.g. the longest-lived isotope of $Z = 105$ has $t_{½} \sim 34$ s. Thus, research on the production of short-lived radionuclides has been particularly intense for the lightest and heaviest elements as well as for the fission products . It is obvious that studies and uses of these short-lived nuclides require special techniques, which often are rather costly. A brief description of some of the techniques used follows.

The sequence of procedures necessary for the production and study of very short-lived nuclei are:

**1)** target preparation and irradiation,
**2)** primary (preliminary) target purification (at irradiation site),
**3)** transport of product (e.g. through pneumatic tubes in which the transport velocity may be up to 100 m $s^{-1}$) from irradiation site to nuclear chemistry laboratory,
**4)** isolation of desired product and eventual labeling, and
**5)** final use in physico-chemical experiments or in applied areas; e.g. hospitals or industry.

A primary requirement for work with short-lived nuclides is that the site or production and the site of use be close together. In 2–30 seconds it is possible to use rapid separation procedures consisting of (a) target dissolution, followed by (b) chemical isolation of the radionuclide through solvent-extraction-stripping or high pressure/temperature ion exchange, and/or (c) precipitation or electro-plating before (d) measurement. In this scheme eventual chemical studies can be included, usually between steps (b) and (c).

There are a number of non-chemical techniques involving time-of-flight and magnetic mass analysis of target recoil products, energy sensitive detectors, etc., with which half-lives down to $10^{-12}$ s can be measured. However, of more interest to us are the fast chemical systems. Let us review a few representative types.

### 17.7.1. Production of $^{11}$C labeled compounds

In §18.5.3 (c) we describe the use of the PET medical imaging technique, which requires the use of positron emitters. Many interesting positron emitters are very short-lived (see Table 18.5) requiring rapid synthesis of the labeled compound. One of the more frequently used positron emitters is $^{11}$C ($t_{½}$ 20.3 m) because it can be introduced into almost any organic compound. Glucose derivatives are of special interest because glucose is the main energy source of the brain; as a consequence, the areas in which glucose accumulates indicate places of highest physiological activity. Also, glucose derivatives are relatively easy to synthesize. To illustrate the technique, consider the synthesis of 2-deoxy-D-[1-$^{11}$C] glucose ($^{11}$C-2DG) which is used to study tumor metabolism in the brain (steps 1 and 2 in §18.5.3 (c)):

By irradiating a target consisting of a mixture of 95% $N_2$ + 5% $H_2$ in a cylinder, 400 mm long and 60 mm in diameter, at 10 bar pressure by 20 MeV protons, $^{11}CH_4$ and $^{13}N_2$ are produced. After an irradiation of 30-60 minutes, the gas is mixed with ammonia and allowed to react at elevated temperature on a platinum catalyst to produce H$^{11}$CN (~80% conversion at 850 °C). The yield is about 5 GBq per μA of proton current. The subsequent synthesis is a series of elaborate steps in which Na$^{11}$CN is digested with preformed reactive glucose derivatives in presence of Ni/Al-alloy in formic acid. The reaction mixture is purified through several ion exchange columns and finally collected on an anion exchanger. The total synthesis, from end of irradiation, takes about 1 hour with an over-all yield of ~80% relative to the H$^{11}$CN yield.

Many procedures for rapid $^{11}$C-labelling are described in the literature. In some cases labeled compounds can be ordered from radiopharmaceutical companies, but for the $^{11}$C-compounds the $t_{½}$ mandates production and synthesis at the place of use.

### 17.7.2. Auto-batch procedures

Figure 17.4 shows an auto-batch procedure for isolation and separation of rare earths obtained from spontaneous fission of $^{252}$Cf. The rare earths plus other fission products are washed from the target chamber by nitric acid and extracted by high-pressure liquid chromatography onto a column containing dihexyl diethyl carbamoyl-methylphosphonate (DHDECMP). After removal from this column, the rare earths are sorbed on a cation exchange column, from which they are eluted sequentially by hot (95°C) α-hydroxy isobutyric acid (HIBA). Figure 17.4 shows a typical elution curve. The total separation procedure takes about 10 minutes.

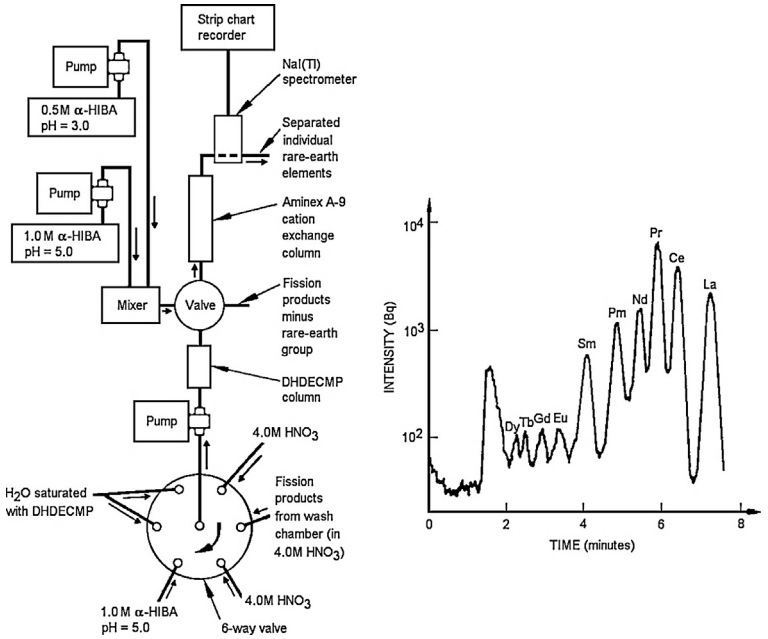



**Figure 17.4** Schematic of auto-batch lanthanide chemistry system and elution curve obtained. *(From Baker et al.)*

Figure 17.5 shows a much more rapid procedure to recover zirconium isotopes formed by fission. (a) A solution of $^{235}U$ sealed in a plastic capsule and irradiated with thermal neutrons is projected pneumatically into the apparatus where the capsule is crushed. (b) The solution is filtered through two layers of preformed, inactive silver chloride, which, by heterogeneous exchange, rapidly removes I and Br. (c) The next step involves solvent extraction with an organic phase containing the reagent tri-butyl phosphate (TBP) adsorbed on the surface of small plastic grains. The filtrate from (b) is mixed with strong $HNO_3 + KBrO_3$ and drawn through the stationary organic phase (marked "movable filter" in the Figure), to remove zirconium. (d) After washing the organic phase with nitric acid, the filter layer containing the zirconium is transferred pneumatically to the detector (to the right in the Figure). (e) Counting is started 4 s after the end of irradiation. The Figure shows the time sequences for the operations in numbers. Stopcocks are operated pneumatically and valves magnetically by signals from electronic timers. The chemical yield of zirconium amounts to about 25%. This experiment led to the identification of $^{99}Zr$ with a half-life of 1.8 s. The ARCA (Automatic Rapid Chemistry Apparatus) in Darmstadt is a modification of the system in Figure 17.5; because the separations are done by ion exchange resins, it is considerably slower, requiring nuclides with $t_{½} > 20$ s.

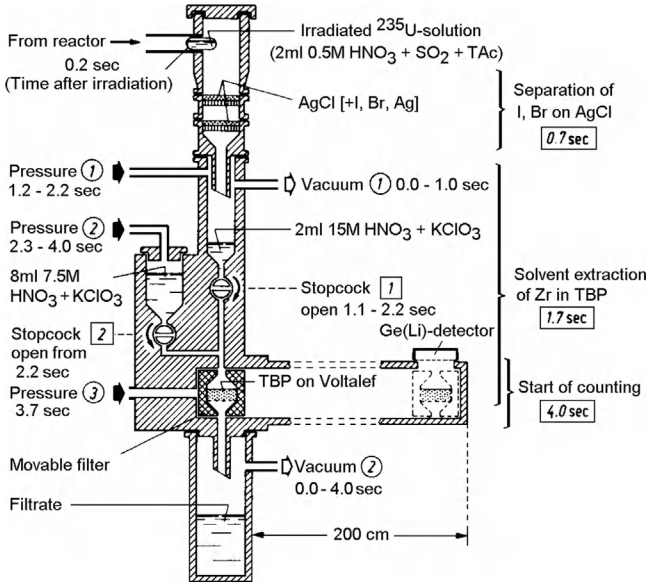



**Figure 17.5** Auto-batch system for identification of short-lived zirconium isotopes. *(From Trautmann et al.)*

In these systems proper modification of conventional techniques provided very fast separations. Precipitation (using a preformed precipitate and heterogeneous exchange), solvent extraction (by reversed phase chromatography), and ion exchange (with heating and pressure) have been used in the identification and study of short-lived transuranides (elements 102 and higher).

## 17.7.3. On-line procedures

For the production of shortlived radioisotopes and for the chemical studies of the heaviest elements it is necessary to have fast chemical systems. In case of the heavy elements this is because the half-lives of most known isotopes of these elements are normally short. In the following we will briefly describe a few such systems and procedures.

### *17.7.3.1. Gas phase separation*

Recoil techniques (§17.6) have been used extensively with *gas-jet transportation systems* for studying short lived radionuclides. As an example, consider the formation of element 103, lawrencium. The $_{103}$Lr nuclides, which were formed in the reaction in the target between $^{252}$Cf and projectiles of $^{11}$B, recoiled from the thin target into the helium gas where they were stopped by atomic collisions with He-atoms. These recoil species became electrically charged cations as the result of loss of atomic electrons in the collision

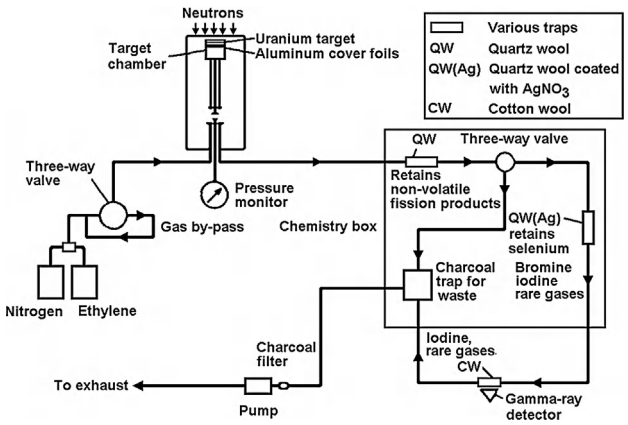



**Figure 17.6** On-line gas-phase separation system for isolation of bromine products. *(From Rengan et al.)*

with the gas atoms. They could, therefore, be attracted to a moving, negatively charged, metal coated plastic band, which carried the Lr atoms on its surface to $\alpha$-particle detectors where the $\alpha$-decay of the lawrencium was measured; see Fig. 17.3. Instead of collecting the particles on a moving band, they could also be swept by a gas or aerosol stream into an apparatus for chemical processing.

Separation may be achieved by having a target containing a reactive compound or having the recoiling species continue in the gas phase by using either a reactive gas or using gas-solid separation. Figure 17.6 illustrates the separation of bromine from other fission products by using gas-solid reactions on quartz and cotton wool. Figure 17.7 illustrates another technique: a chemically reactive thermo-chromatographic column is placed after the target cage. The target $N_2$-gas contains a KCl-aerosol to which the fission product nuclides attach. The gas-aerosol stream passes through a tube or column containing a partly selective adsorbent (e.g. quartz powder, KCl, $BaCl_2$, $K_2CrO_7$). The temperature gradient along the tube leads to deposition at different positions of elements (or compounds) of different vapor pressure. Rather selective deposition can be achieved, although the technique is not totally element specific. Nuclides with half-lives down to 0.1 s have been determined in identification of spallation products, fission products, and transuranium elements. An example is the OLGA system (On-line Gas Chemistry Apparatus) in Darmstadt which was used to identify $^{263}$Ha (atomic number 105); see Ch. 14.

### *17.7.3.2. Gas phase chemistry*

Gas phase chemistry has been very successful for the production of SHE and so far this technique has been used to make Rf, Db, Sg, Bh, Hs and Cn. The original version, OLGA (On-Line Gas chemistry Apparatus) consisted of a thermocromatographic column connected to a reaction oven where the products delivered by the gas jet transport from the target were mixed with reactive gases to form compounds that were fed into the

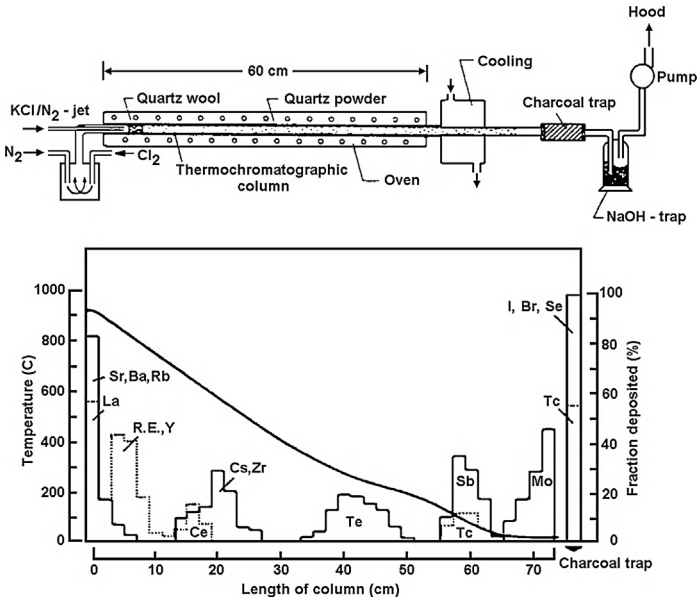



**Figure 17.7** On-line thermo-chromatographic system. *(From Hickmann et al.)*

chromatography system. Detection was achieved by alpha-alpha correlations. This technique has been improved and modified for the production of Bh, Hs, and Cn.

### *17.7.3.3. Solid/liquid extraction and ion exchange*

The ARCA (Automated Rapid Chemistry Apparatus) was developed at the Gesellschaft für Schwerionenforschung (GSI), Darmstadt (Germany). ARCA is an HPLC system based on fast repetitive chromatographic separations in micro columns. The columns are 8 mm long with an inner diameter of 1.6 mm and a dead volume of only ~35 µl. Depending on the separation procedure to be performed, the columns can be filled either with an organic extractant adsorbed on an inert supporting material, or with ion exchanger.

The latest ARCA version, ARCA II, consists of three main sections: a section for collecting activity from the gas-jet from the target and for carrying out the chemical separations, an HPLC section consisting of pumps, valves and solvent reservoirs, an electropneumatic valve system and a control unit, see Figure 17.8. All parts in contact with solutions are made of inert materials like Kel-F, PTFE (polytetrafluoro ethylene) or sapphire. The ARCA system is symmetrical and operated in an alternating mode between left and right. At the same time previously collected activity is dissolved and transported to a column for chemical separation on one side, activity from the gas-jet is collected on a polyethylene frit, and the next column is washed and conditioned on the other side. The products of interest are eluted from the columns onto preheated tantalum

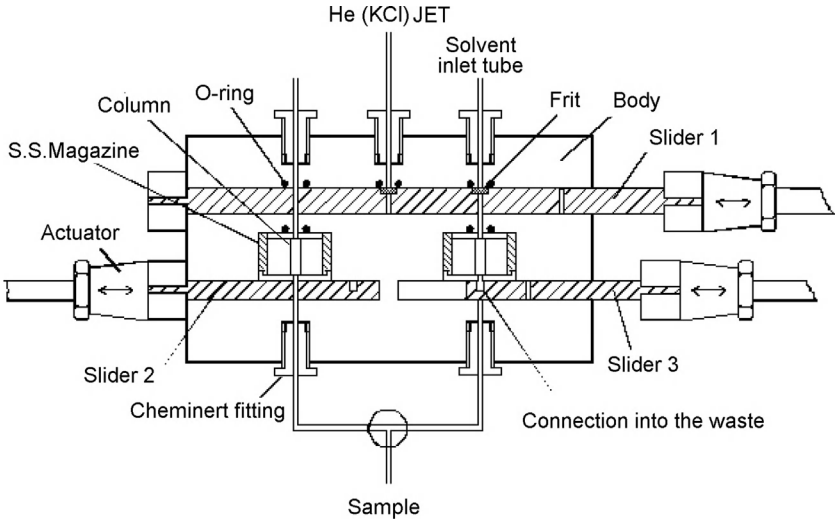



**Figure 17.8** A sketch of the general principle of the main chemical separation section of ARCA II.

or titanium disks and the eluent is evaporated. After flaming and cooling the disk is placed in the measuring position. Typical times from the end of irradiation to the start of measurements is 50 s, which makes it possible to study nuclides with a half-life down to 10−15 s.

#### *17.7.3.4. Solvent extraction*

The SISAK (Short-lived Isotopes Studied by the AKUFVE) technique is illustrated in Figure 17.9. A target containing an element (e.g. uranium) on a thin foil is bombarded with projectiles, while a gas stream containing "catcher" particles (e.g. KCl) behind the target continually removes the products recoiling into the gas stream. The gas jet is thoroughly mixed with an aqueous solution in the static mixer and then degassed. The aqueous solution containing the radioactive products is passed through a series of fast and selective solvent extraction loops, where continuous flow mixers and centrifugal separators are used (the AKUFVE-technique, §18.4.3). The solution from the final separation passes through a "hold-up" cell, which may contain also some absorbent to capture the element of interest. The cell is viewed by a detector system. The procedure can provide high selectivity for a particular element and takes about 3 s from target to the detection site. Because it is an on-line technique, SISAK has mainly been used mainly to obtain detailed decay schemes of very short lived nuclides ($t_{½}$ ~1 to 10 s). It has also been used to study the extraction behaviour of superheavy elements. In this application it is connected to a gas-filled mass separator to reduce interfering background activities. In the newest version, SISAK 3, centrifuges with a hold-up volume of ~0.3 ml/phase and a highest flow rate of 3 ml/s are used.

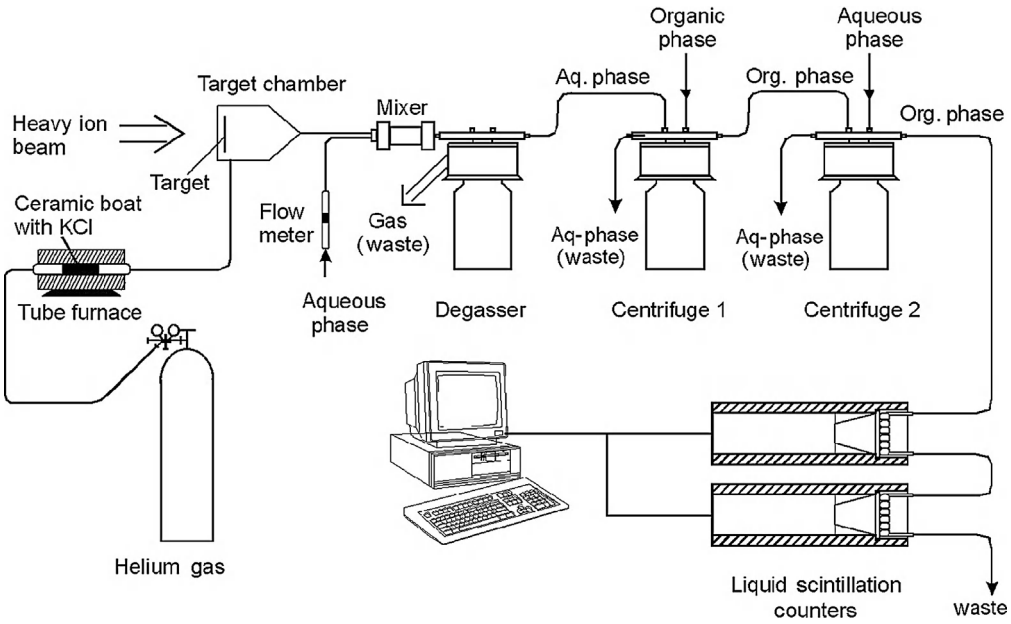



**Figure 17.9** The SISAK fast solvent extraction system. *(Courtesy Skarnemark)*

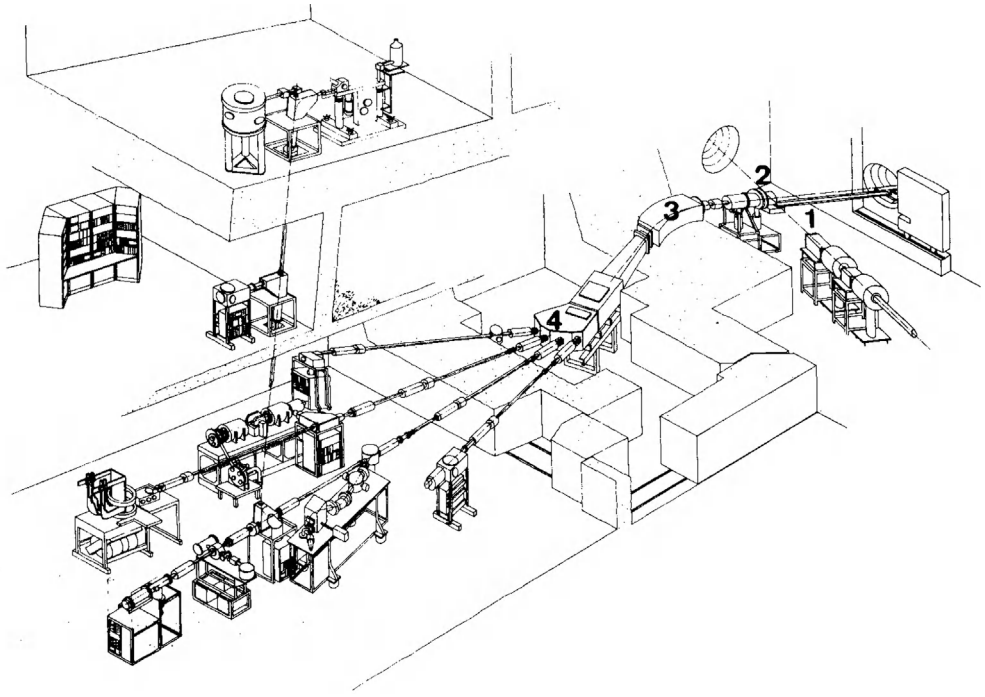



**Figure 17.10** The ISOLDE separator system at CERN. *(Courtesy Rudstam)*

### 17.7.4. Mass separator procedures

In the ISOLDE (Isotope Separation On Line DEvelopment) collaboration at CERN, the target material is fixed in a solid but porous chemical matrix and irradiated by protons or heavy ions. The products react with the target matrix, and those with the "right" chemistry diffuse from the target (which may be heated to increase the diffusion rate) and are swept by a gas stream into the ionization chamber of a mass spectrometer where the products are separated. Figure 17.9 shows the schematic for such a system: At 1, a 600 MeV proton beam is focused on the target 2. The beam of spallation products is accelerated to 60 keV, focused by magnet 3 and separated by electrostatic deflectors at 4 to yield a number of beams which are analyzed by various techniques (nuclear spectroscopy, laser spectroscopy, atomic beam magnetic resonance, range measurements, etc).

## 17.8. EXERCISES

**17.1.** $^{24}$Na is produced through the reaction $^{26}$Mg(d,$\alpha$)$^{24}$Na. A 0.2 mm thick magnesium foil is irradiated for 1 h by a current of 130 μA of 22 MeV $D^+$ ions in a cyclotron. The foil has a much larger area than the cross-section of the beam. What is the specific activity of $^{24}$Na if the magnesium foil (3 cm$^2$) contains 0.003% Na and $\sigma$ for the reaction is assumed to be 25 mb?

**17.2.** Oxygen can be determined through the reaction $^{16}$O(n,p)$^{16}$N $\rightarrow$ ($\beta$,$\gamma$ 7 s); $\sigma_{np}$ at 14 MeV n is 49 mb. 3.982 g of a fatty acid were irradiated for 20 s in $4 \times 10^{12}$ n m$^{-2}$ s$^{-1}$. After the irradiation the sample was rapidly transferred with a rabbit system to a scintillation detector which had an efficiency of 1.1% for the $^{16}$N $\gamma$-rays ($\sim$6 MeV). Exactly 8 s after the end of the irradiation, the sample was counted for 1 min, yielding 13 418 counts above background. What was the oxygen fraction of the sample?

**17.3.** $^{199}$Au can be formed through two successive n,$\gamma$-reactions on $^{197}$Au (100% in nature). If 1 g $^{197}$Au is irradiated with $10^{18}$ n m$^{-2}$ s$^{-1}$ during 30 h, what will the disintegration rate of $^{199}$Au be at the end of the irradiation? The chain of events to be considered is:

$$^{197}\text{Au}\ (\sigma_{n,\gamma}\ 98.8\ \text{b}) \rightarrow {}^{198}\text{Au}\left(t_{1/2}\ 2.694\ \text{d},\ \sigma_{n,\gamma}\ 25\,100\ \text{b}\right) \rightarrow$$
$$^{199}\text{Au}\left(t_{1/2}\ 3.139\ \text{d},\ \sigma_{n,\gamma}\ 30\ \text{b}\right) \rightarrow \text{Neglect self shielding.}$$

**17.4.** $^{246}$Cm has a half-life of 4 730 y. It can be obtained through neutron capture in $^{245}$Cm, which has a half-life of 8 500 y; the reaction cross-section is 345 b. Both isotopes are also fissioned by thermal neutrons, $\sigma_{245}$ 2 020 b and $\sigma_{246}$ 0.17 b.